

Sampling Clock Error Estimation Using a Frequency Reference

1 Sampling-Clock Estimation and Frequency Compensation

1.1 Introduction

A digital frequency-measurement system converts a discrete-time frequency into physical units by multiplying it by an assumed sampling frequency. If the actual sampling clock differs from the nominal clock used by the signal-processing algorithm, all reported frequencies contain a common multiplicative error.

A related correction principle is presented by Kusters, Fischer, and Leach for a digitally compensated counter time base. In their approach, the oscillator frequency is not necessarily stabilized directly. Instead, the time-base error is estimated and a numerical compensation factor is applied to the measured frequency [1]. The relative measurement error therefore depends on the time-base error rather than on the frequency of the measured signal.

The same principle is applicable to the FPGA-based frequency-tracking system. The actual FPGA sampling frequency is estimated using a precision reference signal, after which the resulting correction factor is applied to the measured mode frequencies and to the numerically controlled oscillator.

1.2 Definitions

Let

$$F_0 = \text{nominal FPGA sampling frequency assumed by the digital system,} \quad (1)$$

$$F = \text{actual FPGA sampling frequency,} \quad (2)$$

$$f = \text{true analog input frequency,} \quad (3)$$

$$f_{\text{meas}} = \text{frequency reported using the nominal sampling frequency,} \quad (4)$$

$$f_{\text{LO,cmd}} = \text{local-oscillator frequency commanded under the assumption } F = F_0, \quad (5)$$

$$f_{\text{LO,act}} = \text{actual physical local-oscillator frequency,} \quad (6)$$

$$f_{\text{BB,meas}} = \text{baseband frequency reported using the nominal clock,} \quad (7)$$

$$f_{\text{BB,act}} = \text{actual physical baseband frequency.} \quad (8)$$

For the implemented system,

$$F_0 = 65 \text{ MHz,} \quad (9)$$

$$f_{\text{ref}} = 10 \text{ MHz.} \quad (10)$$

The nominal downsampled sampling frequency is

$$F_{\text{DS},0} = \frac{F_0}{N}, \quad (11)$$

where N is the downsampling factor. For

$$N = 6500, \quad (12)$$

the nominal downsampled frequency is

$$F_{\text{DS},0} = 10 \text{ kHz}. \quad (13)$$

1.3 Frequency Scaling Caused by Sampling-Clock Error

A sinusoidal analog input sampled by the actual clock F is represented as

$$x[n] = \sin\left(2\pi \frac{f}{F} n\right). \quad (14)$$

Its normalized discrete-time frequency is

$$\rho = \frac{f}{F}. \quad (15)$$

The frequency-estimation algorithm converts the normalized frequency into hertz using the nominal clock F_0 . The reported frequency is therefore

$$f_{\text{meas}} = \rho F_0 = f \frac{F_0}{F}. \quad (16)$$

Equation (16) shows that the measured frequency is scaled by the ratio F_0/F . Consequently,

$$F > F_0 \implies f_{\text{meas}} < f, \quad (17)$$

$$F < F_0 \implies f_{\text{meas}} > f. \quad (18)$$

The corresponding frequency error is

$$\Delta f = f_{\text{meas}} - f = f \left(\frac{F_0}{F} - 1 \right). \quad (19)$$

1.4 Fractional Clock Error and Compensation Coefficient

The fractional sampling-clock error is defined as

$$\varepsilon = \frac{F - F_0}{F_0}. \quad (20)$$

It follows that

$$F = F_0(1 + \varepsilon). \quad (21)$$

Substituting this expression into Equation (16) gives

$$f_{\text{meas}} = \frac{f}{1 + \varepsilon}. \quad (22)$$

The exact correction coefficient required to recover the true frequency is

$$C_F = \frac{F}{F_0} = 1 + \varepsilon. \quad (23)$$

Therefore,

$$f = C_F f_{\text{meas}} = f_{\text{meas}} \frac{F}{F_0}. \quad (24)$$

This expression is equivalent to the inverse compensation used for the imperfect counter time base in [1]. If the relative measurement error is written as

$$c = \frac{f_{\text{meas}} - f}{f}, \quad (25)$$

then

$$1 + c = \frac{F_0}{F}, \quad (26)$$

and the corrected frequency becomes

$$f = f_{\text{meas}}(1 + c)^{-1}. \quad (27)$$

Thus, the paper's inverse compensation factor and the FPGA clock-ratio correction are mathematically equivalent.

1.5 First-Order Taylor Approximation

For small clock errors, the reciprocal term in Equation (22) can be expanded as a Taylor series about $\varepsilon = 0$:

$$\frac{1}{1 + \varepsilon} = 1 - \varepsilon + \varepsilon^2 - \varepsilon^3 + \dots, \quad |\varepsilon| < 1. \quad (28)$$

Retaining the constant and first-order terms gives the first-order linear approximation

$$\frac{1}{1 + \varepsilon} \approx 1 - \varepsilon. \quad (29)$$

Consequently,

$$f_{\text{meas}} \approx f(1 - \varepsilon). \quad (30)$$

The approximation therefore shows that the relative measured-frequency error is approximately the negative of the relative sampling-clock error:

$$\frac{f_{\text{meas}} - f}{f} \approx -\varepsilon. \quad (31)$$

The first neglected Taylor-series term is

$$\varepsilon^2. \quad (32)$$

The absolute frequency error introduced by the linearization is therefore approximately

$$e_{\text{lin}} \approx f\varepsilon^2. \quad (33)$$

More precisely, the difference between the exact reciprocal and its first-order approximation is

$$e_{\text{rec}} = \frac{1}{1 + \varepsilon} - (1 - \varepsilon) = \frac{\varepsilon^2}{1 + \varepsilon}. \quad (34)$$

Accordingly, the exact absolute frequency error caused by the first-order approximation is

$$e_{\text{lin}} = f \frac{\varepsilon^2}{1 + \varepsilon}. \quad (35)$$

For a sampling-clock error of

$$\varepsilon = 30 \times 10^{-6}, \quad (36)$$

the first neglected term is

$$\varepsilon^2 = 9 \times 10^{-10}. \quad (37)$$

At an input frequency of 10 MHz, this corresponds to approximately

$$e_{\text{lin}} \approx 10^7 \cdot 9 \times 10^{-10} = 9 \times 10^{-3} \text{ Hz}. \quad (38)$$

The first-order approximation is therefore sufficiently accurate for ppm-level clock errors. Nevertheless, the exact correction factor in Equation (24) can be used when division or a precomputed multiplier is available.

1.6 Estimation of the Actual Sampling Clock

If the true reference frequency f_{ref} is known and its measured value is $f_{\text{ref,meas}}$, then

$$f_{\text{ref,meas}} = f_{\text{ref}} \frac{F_0}{F}. \quad (39)$$

Solving for the actual FPGA sampling frequency gives

$$F = F_0 \frac{f_{\text{ref}}}{f_{\text{ref,meas}}}. \quad (40)$$

Using the definition of the fractional clock error,

$$\varepsilon = \frac{F - F_0}{F_0}, \quad (41)$$

it follows that

$$\frac{F}{F_0} = 1 + \varepsilon. \quad (42)$$

Substituting Equation (40) gives

$$1 + \varepsilon = \frac{f_{\text{ref}}}{f_{\text{ref,meas}}}. \quad (43)$$

Therefore, the fractional sampling-clock error can be obtained directly from the reference-frequency measurement as

$$\boxed{\varepsilon = \frac{f_{\text{ref}}}{f_{\text{ref,meas}}} - 1}. \quad (44)$$

The sampling-clock error expressed in parts per million is then

$$\boxed{\text{ppm}_F = 10^6 \varepsilon}. \quad (45)$$

Equivalently,

$$\text{ppm}_F = 10^6 \left(\frac{f_{\text{ref}}}{f_{\text{ref,meas}}} - 1 \right). \quad (46)$$

1.7 Numerically Controlled Local Oscillator

The digital local oscillator is generated by a phase accumulator or DDS driven by the same FPGA clock F . Let P denote the accumulator width and let $\Delta\phi$ denote the phase increment.

When the phase increment is calculated using the nominal clock,

$$\Delta\phi = \frac{f_{\text{LO,cmd}}}{F_0} 2^P. \quad (47)$$

The actual physical DDS output frequency is

$$f_{\text{LO,act}} = \frac{\Delta\phi}{2^P} F. \quad (48)$$

Substitution of Equation (47) gives

$$f_{\text{LO,act}} = f_{\text{LO,cmd}} \frac{F}{F_0}. \quad (49)$$

The DDS local oscillator therefore contains the same fractional error as the FPGA sampling clock. A faster FPGA clock produces a proportionally higher physical LO frequency, whereas a slower FPGA clock produces a proportionally lower physical LO frequency.

1.8 Digital Downconversion Model

The normalized input frequency is

$$\rho_{\text{in}} = \frac{f}{F}. \quad (50)$$

The normalized DDS frequency is determined by the programmed phase increment:

$$\rho_{\text{LO}} = \frac{\Delta\phi}{2^P} = \frac{f_{\text{LO,cmd}}}{F_0}. \quad (51)$$

After complex mixing, the normalized baseband frequency is

$$\rho_{\text{BB}} = \rho_{\text{in}} - \rho_{\text{LO}} = \frac{f}{F} - \frac{f_{\text{LO,cmd}}}{F_0}. \quad (52)$$

The FFT converts this normalized baseband frequency into hertz using the nominal clock. Therefore,

$$f_{\text{BB,meas}} = F_0 \rho_{\text{BB}}. \quad (53)$$

The measured baseband frequency is consequently

$$f_{\text{BB,meas}} = f \frac{F_0}{F} - f_{\text{LO,cmd}}. \quad (54)$$

Equation (54) already includes the dependence of the DDS local oscillator on the actual FPGA clock. The commanded LO frequency $f_{\text{LO,cmd}}$ is expressed in nominal-clock hertz, whereas the actual physical LO frequency is given by Equation (49).

Rearranging Equation (54) gives

$$f \frac{F_0}{F} = f_{\text{LO,cmd}} + f_{\text{BB,meas}}. \quad (55)$$

The true analog input frequency is therefore

$$f = (f_{\text{LO,cmd}} + f_{\text{BB,meas}}) \frac{F}{F_0}. \quad (56)$$

Thus, the complete reconstructed frequency, consisting of the commanded LO and the measured baseband residual, must be multiplied by the clock-correction factor F/F_0 .

1.9 Sampling-Clock Estimation from a Baseband Reference

For a known reference input f_{ref} , Equation (54) becomes

$$f_{\text{BB,meas}} = f_{\text{ref}} \frac{F_0}{F} - f_{\text{LO,cmd}}. \quad (57)$$

Solving for F yields

$$F = F_0 \frac{f_{\text{ref}}}{f_{\text{LO,cmd}} + f_{\text{BB,meas}}}. \quad (58)$$

The apparent reference frequency reconstructed in nominal-clock units is

$$f_{\text{ref,meas}} = f_{\text{LO,cmd}} + f_{\text{BB,meas}}. \quad (59)$$

Equation (58) can therefore also be written as

$$F = F_0 \frac{f_{\text{ref}}}{f_{\text{ref,meas}}}. \quad (60)$$

1.10 Numerical Example

Let

$$F_0 = 65\,000\,000 \text{ Hz}, \quad (61)$$

$$f_{\text{ref}} = 10\,000\,000 \text{ Hz}, \quad (62)$$

$$f_{\text{LO,cmd}} = 9\,999\,000 \text{ Hz}, \quad (63)$$

$$f_{\text{BB,meas}} = 717 \text{ Hz}. \quad (64)$$

The apparent measured reference frequency is

$$f_{\text{ref,meas}} = 9\,999\,000 + 717 = 9\,999\,717 \text{ Hz}. \quad (65)$$

The estimated FPGA clock is therefore

$$F = 65\,000\,000 \frac{10\,000\,000}{9\,999\,717} = 65\,001\,839.6 \text{ Hz}. \quad (66)$$

The clock correction coefficient is

$$C_F = \frac{F}{F_0} = 1.0000283015. \quad (67)$$

The clock error is

$$\Delta F = F - F_0 = 1839.6 \text{ Hz}, \quad (68)$$

$$\text{ppm}_F = 28.3 \text{ ppm}. \quad (69)$$

The actual physical DDS local-oscillator frequency is

$$f_{\text{LO,act}} = 9\,999\,000 \frac{65\,001\,839.6}{65\,000\,000} \approx 9\,999\,283 \text{ Hz}. \quad (70)$$

The corrected input frequency is

$$f = 9\,999\,717 \frac{65\,001\,839.6}{65\,000\,000} \approx 10\,000\,000 \text{ Hz}. \quad (71)$$

The physical baseband difference is

$$f_{\text{BB,act}} = f - f_{\text{LO,act}}. \quad (72)$$

Equivalently,

$$f_{\text{BB,act}} = f_{\text{BB,meas}} \frac{F}{F_0}. \quad (73)$$

For the numerical example,

$$f_{\text{BB,act}} = 717 \frac{65\,001\,839.6}{65\,000\,000} \approx 717.02 \text{ Hz}. \quad (74)$$

1.11 Correction of Tracked Mode Frequencies

Let the tracked mode-frequency vector be

$$\mathbf{M}_m = \begin{bmatrix} M_{m,1} \\ M_{m,2} \\ M_{m,3} \end{bmatrix}, \quad (75)$$

where each element is reconstructed from the corresponding commanded LO and measured baseband residual:

$$M_{m,i} = f_{\text{LO,cmd},i} + f_{\text{BB,meas},i}. \quad (76)$$

Because all channels use the same FPGA sampling clock, the same correction coefficient applies to every element. The corrected vector is

$$\mathbf{M}_{\text{corr}} = \frac{F}{F_0} \mathbf{M}_m. \quad (77)$$

Explicitly,

$$\mathbf{M}_{\text{corr}} = \begin{bmatrix} M_{m,1} \frac{F}{F_0} \\ M_{m,2} \frac{F}{F_0} \\ M_{m,3} \frac{F}{F_0} \end{bmatrix}. \quad (78)$$

The complete reconstructed mode frequency must be corrected. Correcting only the baseband residual while leaving the commanded LO unchanged would be incorrect because the physical DDS frequency is also proportional to the actual FPGA clock.

1.12 Correction of the Downsampled FFT Frequency Axis

Since the downsample enable is generated by dividing the actual FPGA clock by N , the true downsampled sampling frequency is

$$F_{\text{DS}} = \frac{F}{N}. \quad (79)$$

For $N = 6500$,

$$F_{\text{DS}} = \frac{F}{6500}. \quad (80)$$

Using the estimated clock from the numerical example,

$$F_{\text{DS}} = \frac{65\,001\,839.6}{6500} = 10\,000.283 \text{ Hz.} \quad (81)$$

The FFT frequency axis can therefore be corrected directly by using F_{DS} rather than the nominal value 10 000 Hz. This produces the same multiplicative correction as Equation (77).

1.13 Correction of the DDS Phase Increment

When the objective is to generate a specified physical LO frequency rather than merely correct a measured value, the phase increment should be calculated using the estimated actual FPGA clock:

$$\Delta\phi_{\text{corr}} = \frac{f_{\text{LO,desired}}}{F} 2^P. \quad (82)$$

If the phase-increment interface is formulated in terms of an equivalent frequency command referenced to F_0 , the corrected command is

$$f_{\text{LO,cmd,corr}} = f_{\text{LO,desired}} \frac{F_0}{F}. \quad (83)$$

The resulting physical DDS frequency is then

$$f_{\text{LO,act}} = f_{\text{LO,cmd,corr}} \frac{F}{F_0} = f_{\text{LO,desired}}. \quad (84)$$

Thus, the estimated sampling clock can be used both to correct measured frequencies and to periodically update the DDS phase increment.

1.14 Summary

The principal relationships are

$$f_{\text{meas}} = f \frac{F_0}{F}, \quad (85)$$

$$C_F = \frac{F}{F_0}, \quad (86)$$

$$f = C_F f_{\text{meas}}, \quad (87)$$

$$f_{\text{LO,act}} = f_{\text{LO,cmd}} \frac{F}{F_0}, \quad (88)$$

$$f_{\text{BB,meas}} = f \frac{F_0}{F} - f_{\text{LO,cmd}}, \quad (89)$$

$$f = (f_{\text{LO,cmd}} + f_{\text{BB,meas}}) \frac{F}{F_0}, \quad (90)$$

$$F = F_0 \frac{f_{\text{ref}}}{f_{\text{LO,cmd}} + f_{\text{BB,meas}}}, \quad (91)$$

$$\mathbf{M}_{\text{corr}} = \frac{F}{F_0} \mathbf{M}_m, \quad (92)$$

$$F_{\text{DS}} = \frac{F}{6500}, \quad (93)$$

$$\Delta\phi_{\text{corr}} = \frac{f_{\text{LO,desired}}}{F} 2^P. \quad (94)$$

The method follows the same general compensation principle as the digitally compensated time base described in [1]: an error reference is measured, a correction coefficient is obtained, and

the coefficient is applied to recover the corrected frequency. In the FPGA implementation, the precision reference provides a direct estimate of the actual sampling clock, enabling correction of the FFT frequency axis, tracked mode frequencies, and DDS phase increment.

1.15 Statistical Temperature Model

The relative frequency deviation of each resonant mode is assumed to vary linearly with the temperature difference relative to a reference temperature. For mode i , the measurement model is

$$p_i = \alpha_i d + e_i, \quad (95)$$

where

- p_i is the measured frequency deviation of mode i , expressed in ppm,
- α_i is the experimentally determined linear temperature coefficient of mode i , expressed in ppm/°C,
- $d = T - T_0$ is the temperature difference relative to the reference temperature T_0 ,
- e_i is the residual error associated with the measurement.

Throughout this work, the reference temperature is defined as

$$T_0 = 25^\circ\text{C}. \quad (96)$$

The temperature-coefficient vector is

$$\boldsymbol{\alpha} = \begin{bmatrix} -0.359 \\ -34.576 \\ -1.191 \end{bmatrix} \frac{\text{ppm}}{^\circ\text{C}}. \quad (97)$$

The three resonant modes originate from the same quartz resonator and are therefore assumed to experience the same physical temperature. Consequently, the temperature estimation problem contains a single unknown temperature difference d , while each resonant mode provides an independent observation of that quantity.

Collecting the three measurement equations yields the vector model

$$\mathbf{p} = \boldsymbol{\alpha} d + \mathbf{e}, \quad (98)$$

where

$$\mathbf{p} = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}, \quad \mathbf{e} = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}. \quad (99)$$

Equation (98) represents an overdetermined linear system consisting of three measurement equations and a single unknown parameter. An estimate of the temperature difference d is therefore obtained by solving an optimization problem rather than by directly inverting the individual measurement equations.

1.16 Gaussian Error Model

The residual error vector is modeled as a zero-mean Gaussian random vector,

$$\mathbf{e} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}), \quad (100)$$

where $\mathbf{\Sigma}$ is the covariance matrix of the mode measurement errors.

If the residual errors of the three modes are assumed mutually independent, the covariance matrix is diagonal:

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{bmatrix}. \quad (101)$$

The parameters σ_i represent the standard deviations of the residual mode errors after removal of the modeled temperature dependence. Their values are not assumed to be equal to the [standard deviations](#) calculated over the complete temperature sweep. The latter also contain the deterministic variation of frequency with temperature and therefore do not directly represent measurement uncertainty.

The values of σ_i remain model parameters until they are estimated from repeated measurements at constant temperature or from the residuals of the fitted temperature-frequency models.

Under the Gaussian error assumption, estimation of the unknown temperature difference can be formulated as a maximum-likelihood problem. This optimization is equivalent to minimizing a weighted least-squares cost function.

1.17 Weighted Least-Squares Temperature Estimator

For a known positive-definite covariance matrix $\mathbf{\Sigma}$, the temperature difference is estimated by minimizing the weighted squared residual

$$J(d) = (\mathbf{p} - \boldsymbol{\alpha}d)^T \mathbf{\Sigma}^{-1} (\mathbf{p} - \boldsymbol{\alpha}d). \quad (102)$$

Under the diagonal covariance assumption, this cost function becomes

$$J(d) = \sum_{i=1}^3 \frac{(p_i - \alpha_i d)^2}{\sigma_i^2}. \quad (103)$$

The estimate is obtained by satisfying

$$\frac{dJ}{dd} = 0. \quad (104)$$

Differentiation gives

$$-2\boldsymbol{\alpha}^T \mathbf{\Sigma}^{-1} (\mathbf{p} - \boldsymbol{\alpha}d) = 0. \quad (105)$$

The resulting weighted least-squares estimate is

$$\boxed{\hat{d} = (\boldsymbol{\alpha}^T \mathbf{\Sigma}^{-1} \boldsymbol{\alpha})^{-1} \boldsymbol{\alpha}^T \mathbf{\Sigma}^{-1} \mathbf{p}}. \quad (106)$$

Under the Gaussian error assumption, Equation (106) is also the maximum-likelihood estimate of the temperature difference.

For independent mode errors, the scalar form is

$$\hat{d} = \frac{\sum_{i=1}^3 \frac{\alpha_i p_i}{\sigma_i^2}}{\sum_{i=1}^3 \frac{\alpha_i^2}{\sigma_i^2}}. \quad (107)$$

The estimated absolute temperature is

$$\hat{T} = T_0 + \hat{d}. \quad (108)$$

1.18 Equivalent Estimator Row Vector

For fixed values of the temperature coefficients and covariance parameters, the estimator can be written as

$$\mathbf{W} = (\boldsymbol{\alpha}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\alpha})^{-1} \boldsymbol{\alpha}^T \boldsymbol{\Sigma}^{-1}. \quad (109)$$

The estimator row vector has dimensions

$$\mathbf{W} \in \mathbb{R}^{1 \times 3}, \quad (110)$$

and the temperature estimate becomes

$$\hat{d} = \mathbf{W} \mathbf{p}. \quad (111)$$

Although the final implementation is a dot product, the elements of $\mathbf{W}$ are not obtained by independently inverting the temperature coefficients. They are determined jointly from the temperature sensitivities and the covariance model.

The estimator satisfies

$$\mathbf{W} \boldsymbol{\alpha} = 1. \quad (112)$$

Consequently, in the absence of residual error,

$$\mathbf{p} = \boldsymbol{\alpha} d \quad (113)$$

implies

$$\hat{d} = d. \quad (114)$$

1.19 Uncertainty of the Temperature Estimate

For a known covariance matrix, the variance of the weighted least-squares estimate is

$$\sigma_{\hat{d}}^2 = (\boldsymbol{\alpha}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\alpha})^{-1}. \quad (115)$$

The corresponding standard uncertainty is

$$\sigma_{\hat{d}} = \sqrt{(\boldsymbol{\alpha}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\alpha})^{-1}}. \quad (116)$$

The numerical uncertainty of the temperature estimate depends on the covariance matrix of the residual errors and therefore cannot be determined until the covariance matrix has been identified experimentally.

1.20 Sampling-Clock-Corrected ppm Deviations

The measured mode frequencies are initially reconstructed under the assumption that the FPGA sampling clock is equal to its nominal value

$$F_0 = 65 \text{ MHz}. \quad (117)$$

Let the uncorrected measured-frequency vector be

$$\mathbf{M}_m = \begin{bmatrix} M_{m,1} \\ M_{m,2} \\ M_{m,3} \end{bmatrix}. \quad (118)$$

If the estimated actual FPGA clock is F , the sampling-clock-corrected frequency vector is

$$\mathbf{M}_{\text{corr}} = \frac{F}{F_0} \mathbf{M}_m. \quad (119)$$

The corrected frequency of mode i is therefore

$$M_{\text{corr},i} = M_{m,i} \frac{F}{F_0}. \quad (120)$$

The ppm deviation used by the statistical temperature estimator is

$$p_i = 10^6 \frac{M_{\text{corr},i} - M_{n,i}}{M_{n,i}}. \quad (121)$$

Substituting the clock correction gives

$$p_i = 10^6 \frac{M_{m,i} \frac{F}{F_0} - M_{n,i}}{M_{n,i}}. \quad (122)$$

The clock correction must be applied before the temperature estimate is formed. Otherwise, the ppm deviations contain both the physical temperature dependence of the resonant modes and the common frequency scaling caused by the FPGA sampling-clock error.

1.21 Division-Free ppm Conversion

The run-time division by the nominal mode frequency can be eliminated by defining the pre-computed constants

$$K_i = \frac{10^6}{M_{n,i}}. \quad (123)$$

Their units are

$$[K_i] = \frac{\text{ppm}}{\text{Hz}}. \quad (124)$$

For the three modes,

$$K_1 = 0.09996001599360256 \frac{\text{ppm}}{\text{Hz}}, \quad (125)$$

$$K_2 = 0.15949515387607538 \frac{\text{ppm}}{\text{Hz}}, \quad (126)$$

$$K_3 = 0.295107658213381 \frac{\text{ppm}}{\text{Hz}}. \quad (127)$$

The corrected ppm deviations can then be evaluated using multiplication only:

$$p_1 = \left(M_{m,1} \frac{F}{F_0} - 10.004000 \times 10^6 \right) K_1, \quad (128)$$

$$p_2 = \left(M_{m,2} \frac{F}{F_0} - 6.269781 \times 10^6 \right) K_2, \quad (129)$$

$$p_3 = \left(M_{m,3} \frac{F}{F_0} - 3.388594 \times 10^6 \right) K_3. \quad (130)$$

1.22 Complete Temperature-Estimation Procedure

The complete processing sequence is

tracked frequencies $\longrightarrow$ sampling-clock correction $\longrightarrow$ ppm deviations $\longrightarrow$ weighted least-squares estimate. (131)

For every estimation update, the following operations are performed:

1. Obtain the reconstructed mode frequencies $M_{m,1}$, $M_{m,2}$, and $M_{m,3}$.
2. Correct all three frequencies using the estimated sampling-clock ratio F/F_0 .
3. Subtract the nominal mode frequencies measured at 25 °C.
4. Convert the corrected frequency deviations from hertz to ppm.
5. Form the ppm-deviation vector $\mathbf{p}$.
6. Calculate the temperature difference using

$$\hat{d} = (\boldsymbol{\alpha}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\alpha})^{-1} \boldsymbol{\alpha}^T \boldsymbol{\Sigma}^{-1} \mathbf{p}.$$

7. Calculate the absolute temperature using

$$\hat{T} = 25^\circ\text{C} + \hat{d}.$$

Since all model parameters are constant, the estimator row vector $\mathbf{W}$ can be precomputed. The run-time temperature calculation then reduces to

$$\hat{d} = \mathbf{W} \mathbf{p}. \quad (132)$$

The clock correction removes the common frequency-scaling error before the mode-dependent deviations are statistically combined. The weighted least-squares stage then accounts for the different temperature sensitivities and measurement uncertainties of the three resonant modes.

1.23 Summary

The nominal mode-frequency vector is

$$\mathbf{M}_n = \begin{bmatrix} 10.004000 \times 10^6 \\ 6.269781 \times 10^6 \\ 3.388594 \times 10^6 \end{bmatrix} \text{ Hz}. \quad (133)$$

The sampling-clock-corrected frequency vector is

$$\mathbf{M}_{\text{corr}} = \frac{F}{F_0} \mathbf{M}_m. \quad (134)$$

The corrected ppm-deviation vector contains the elements

$$p_i = 10^6 \frac{M_{\text{corr},i} - M_{n,i}}{M_{n,i}}. \quad (135)$$

The linear statistical model is

$$\mathbf{p} = \boldsymbol{\alpha}d + \mathbf{e}, \quad \mathbf{e} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}). \quad (136)$$

The weighted least-squares temperature estimate is

$$\hat{d} = (\boldsymbol{\alpha}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\alpha})^{-1} \boldsymbol{\alpha}^T \boldsymbol{\Sigma}^{-1} \mathbf{p}. \quad (137)$$

For a known covariance matrix, the weighted least-squares estimator is

$$\hat{d} = \mathbf{W} \mathbf{p}, \quad (138)$$

where

$$\mathbf{W} = (\boldsymbol{\alpha}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\alpha})^{-1} \boldsymbol{\alpha}^T \boldsymbol{\Sigma}^{-1}. \quad (139)$$

Finally, the estimated absolute temperature is

$$\hat{T} = 25^\circ\text{C} + \hat{d}. \quad (140)$$

1.24 Open Research Questions

The proposed temperature-estimation framework establishes the mathematical model required to estimate the resonator temperature from the measured mode frequencies.

1.24.1 Identification of the Residual Covariance Matrix

The weighted least-squares estimator requires knowledge of the covariance matrix of the residual measurement errors,

$$\boldsymbol{\Sigma} = \text{Cov}(\mathbf{e}). \quad (141)$$

Although the mathematical formulation is complete, the covariance matrix has not yet been identified experimentally. Consequently, the weighting matrix of the estimator remains symbolic. How to get real values?

Once the covariance matrix has been identified, the optimal weighting matrix can be pre-computed and incorporated into the FPGA implementation.

1.24.2 Calibration of the Sampling-Clock Error Without a Reference Signal

The sampling-clock estimation method proposed in this work requires the presence of a precision reference signal. During normal operation, however, such a reference signal may not be available. Consequently, an alternative observable quantity is required to estimate the sampling-clock error.

Kusters *et al.* [1] addressed an analogous problem by experimentally characterizing the relationship between the frequency error of the oscillator and an internal measurable quantity, namely the frequency difference $f_B - f_C$. The measured data were then used to construct a calibration function that mapped the observable quantity directly to the oscillator correction coefficient.

The objective of such a calibration is to identify a mapping of the form

$$\varepsilon = g(x), \quad (142)$$

where x denotes the selected observable quantity and ε is the fractional sampling-clock error.

Once the function $g(\cdot)$ has been identified experimentally, the external reference signal is no longer required during normal operation. Instead, the observable quantity x is measured, the sampling-clock error is estimated using the calibrated model, and the corresponding frequency correction coefficient is applied throughout the signal processing chain.

1.25 Results

For the results presented in this section, the reference-frequency estimator used a 64-point FFT. The covariance matrix of the temperature estimator was evaluated under the initial assumption

$$\sigma_1 = \sigma_2 = \sigma_3 = 1 \text{ ppm}. \quad (143)$$

Consequently,

$$\mathbf{\Sigma} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ppm}^2. \quad (144)$$

This choice does not represent experimentally determined residual variances. It assigns equal statistical weighting to all three modes and is used as an initial implementation assumption until the residual covariance matrix can be characterized experimentally.

The frequencies reported for channels 1–3 are corrected using the sampling-clock correction factor

$$C_F = \frac{F}{F_0}. \quad (145)$$

Thus, the reported mode frequencies correspond to

$$M_{\text{corr},i} = C_F M_{m,i}. \quad (146)$$

These corrected frequencies are subsequently used to calculate the relative frequency deviations required by the temperature estimator.

The correction is applied only to the reconstructed frequency measurements. The raw frequency estimates remain in use by the upconversion stage, since the DDS phase increment operates in the FPGA clock domain and must therefore remain consistent with the actual digital signal-processing chain.

A representative sequence of measured results is shown below:

f_1 [Hz]	f_2 [Hz]	f_3 [Hz]	$f_{\text{ref,BB}}$ [Hz]	F [Hz]	$\hat{d}$ [°C]	
10003877.543	6269144.731	3388389.802	1073.034	64999526.000	2.994	
10003858.074	6269131.112	3388384.047	1092.920	64999397.000	3.059	
10003855.900	6269129.517	3388383.686	1093.942	64999390.000	3.067	
10004246.666	6269374.838	3388515.623	701.180	65001943.000	1.886	
10004199.960	6269344.678	3388498.792	747.717	65001640.000	2.032	
10004168.141	6269318.482	3388486.544	778.696	65001439.000	2.157	
10004165.530	6269308.093	3388484.819	781.250	65001422.000	2.205	
10004163.993	6269293.511	3388482.192	782.857	65001412.000	2.273	
10004131.027	6269263.207	3388468.971	815.754	65001198.000	2.418	
10004012.382	6269176.286	3388426.928	934.440	65000427.000	2.834	(147)

The tabulated results demonstrate the simultaneous operation of the reference-based sampling-clock estimator and the three-mode temperature estimator. Variations in the measured reference residual produce corresponding estimates of the actual FPGA sampling frequency F . The resulting sampling-clock correction is applied to the reconstructed mode frequencies before their deviations from the nominal temperature-calibration frequencies are evaluated.

Figure 1 shows the reference-channel residual frequency and the estimated temperature difference d_{temp} over time.

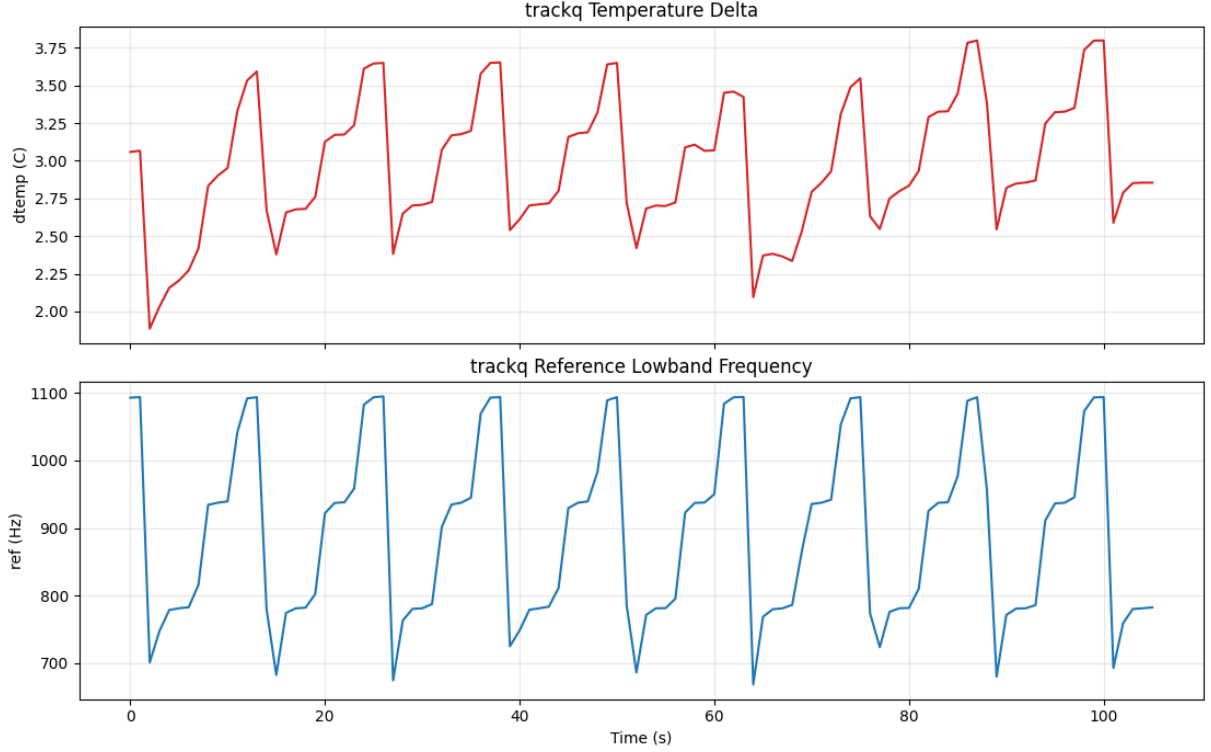


Figure 1: Estimated temperature difference d_{temp} and reference-channel lowband frequency over time.

1.26 Residual Variance Estimation of the Resonant Modes

The covariance matrix used by the temperature estimator requires an estimate of the residual variance of each resonant mode. The variance should not be calculated directly from the frequency variation over the temperature sweep, since this variation contains the deterministic temperature dependence of the resonator. Instead, a linear temperature model is first fitted to each mode and the variance is calculated from the residuals of this model.

For each resonant mode, ten measurements were available in the temperature range from 20°C to 65°C in steps of 5°C. The measured resonant frequency at 25°C was selected as the reference frequency for the ppm conversion.

For each measured frequency f_i , the relative frequency deviation was calculated as

$$p_i = 10^6 \frac{f_i - f_{\text{ref}}}{f_{\text{ref}}}, \quad (148)$$

where p_i is expressed in ppm.

The temperature dependence was approximated by the linear model

$$p_i = a + bT_i + r_i, \quad (149)$$

where a is the intercept, b is the temperature coefficient in ppm/°C, and r_i is the residual of the linear fit.

The parameters a and b were estimated using ordinary least squares. The fitted value is therefore

$$\hat{p}_i = \hat{a} + \hat{b}T_i, \quad (150)$$

and the residual corresponding to each measurement is

$$r_i = p_i - \hat{p}_i = p_i - (\hat{a} + \hat{b}T_i). \quad (151)$$

The residual sum of squares is

$$\text{SSE} = \sum_{i=1}^N r_i^2. \quad (152)$$

Since both the slope and intercept are estimated from the same data, the unbiased estimate of the residual variance is

$$\sigma^2 = \frac{\text{SSE}}{N - 2} \quad (153)$$

with $N = 10$, giving $N - 2 = 8$ residual degrees of freedom.

1.26.1 C100 Mode

For the C100 mode, the measured resonance frequency at 25°C is

$$f_{\text{ref,C100}} = 3\,388\,594 \text{ Hz}. \quad (154)$$

The measured resonance frequencies are

$$\mathbf{f}_{\text{C100}} = [3388594, 3388594, 3388594, 3388594, 3388594, \\ 3388594, 3388594, 3388438, 3388438, 3388438] \text{ Hz}. \quad (155)$$

After conversion to ppm and linear regression, the fitted model is

$$\hat{p}_{\text{C100}}(T) = 35.9924 - 1.171846T \quad (156)$$

and therefore the temperature coefficient is

$$b_{\text{C100}} = -1.171846 \frac{\text{ppm}}{^\circ\text{C}}. \quad (157)$$

The residuals of the fitted model are approximately

$$\mathbf{r}_{\text{C100}} = [-12.5555, -6.6963, -0.8370, 5.0222, 10.8814, \\ 16.7407, 22.5999, -17.5777, -11.7185, -5.8592] \text{ ppm}. \quad (158)$$

The corresponding residual sum of squares is

$$\text{SSE}_{\text{C100}} = 1618.4404 \text{ ppm}^2, \quad (159)$$

which gives

$$\sigma_{\text{C100}}^2 = \frac{1618.4404}{8} = 202.3051 \text{ ppm}^2. \quad (160)$$

The corresponding residual standard deviation is

$$\sigma_{\text{C100}} = 14.2234 \text{ ppm.} \quad (161)$$

1.26.2 B100 Mode

For the B100 mode,

$$f_{\text{ref,B100}} = 3\,727\,969 \text{ Hz.} \quad (162)$$

The measured resonance frequencies are

$$\mathbf{f}_{\text{B100}} = [3728594, 3727969, 3727813, 3726875, 3726094, \\ 3726250, 3725469, 3724688, 3723906, 3723906] \text{ Hz.} \quad (163)$$

The fitted linear model is

$$\boxed{\hat{p}_{\text{B100}}(T) = 761.8756 - 29.366872T} \quad (164)$$

with temperature coefficient

$$\boxed{b_{\text{B100}} = -29.366872 \frac{\text{ppm}}{^\circ\text{C}}.} \quad (165)$$

The residuals are

$$\mathbf{r}_{\text{B100}} = [-6.8865, -27.7038, 77.2848, -27.4924, -90.1555, \\ 98.5247, 35.8616, -26.8015, -89.7328, 57.1015] \text{ ppm.} \quad (166)$$

The residual sum of squares is

$$\text{SSE}_{\text{B100}} = 38695.7591 \text{ ppm}^2, \quad (167)$$

giving

$$\boxed{\sigma_{\text{B100}}^2 = \frac{38695.7591}{8} = 4836.9699 \text{ ppm}^2.} \quad (168)$$

The corresponding residual standard deviation is

$$\sigma_{\text{B100}} = 69.5483 \text{ ppm.} \quad (169)$$

1.26.3 A100 Mode

For the A100 mode,

$$f_{\text{ref,A100}} = 6\,269\,781 \text{ Hz.} \quad (170)$$

The measured resonance frequencies are

$$\mathbf{f}_{\text{A100}} = [6271641, 6269781, 6269281, 6267656, 6266289, \\ 6266000, 6264766, 6263594, 6262617, 6261641] \text{ Hz.} \quad (171)$$

The fitted linear model is

$$\boxed{\hat{p}_{\text{A100}}(T) = 907.7993 - 34.323755T} \quad (172)$$

with temperature coefficient

$$\boxed{b_{A100} = -34.323755 \frac{\text{ppm}}{^{\circ}\text{C}}.} \quad (173)$$

The residuals are

$$\mathbf{r}_{A100} = [75.3368, -49.7055, 42.1657, -45.3952, -91.8064, \\ 33.7183, 8.5199, -6.7897, 9.0023, 24.9538] \text{ ppm}. \quad (174)$$

The residual sum of squares is

$$\text{SSE}_{A100} = 22372.7028 \text{ ppm}^2, \quad (175)$$

and the residual variance is

$$\boxed{\sigma_{A100}^2 = \frac{22372.7028}{8} = 2796.5879 \text{ ppm}^2.} \quad (176)$$

The corresponding residual standard deviation is

$$\sigma_{A100} = 52.8828 \text{ ppm}. \quad (177)$$

1.26.4 C300 Mode

For the C300 mode,

$$f_{\text{ref},C300} = 10\,004\,000 \text{ Hz}. \quad (178)$$

The measured resonance frequencies are

$$\mathbf{f}_{C300} = [10004000, 10004000, 10003990, 10003980, 10003960, \\ 10003950, 10003930, 10003900, 10003870, 10003840] \text{ Hz}. \quad (179)$$

The fitted model is

$$\boxed{\hat{p}_{C300}(T) = 9.44471 - 0.358644T} \quad (180)$$

and therefore

$$\boxed{b_{C300} = -0.358644 \frac{\text{ppm}}{^{\circ}\text{C}}.} \quad (181)$$

The residuals are

$$\mathbf{r}_{C300} = [-2.2718, -0.4786, 0.3150, 1.1086, 0.9027, \\ 1.6963, 1.4903, 0.2847, -0.9208, -2.1264] \text{ ppm}. \quad (182)$$

The residual sum of squares is

$$\text{SSE}_{C300} = 18.08250 \text{ ppm}^2, \quad (183)$$

which gives

$$\boxed{\sigma_{C300}^2 = \frac{18.08250}{8} = 2.260313 \text{ ppm}^2.} \quad (184)$$

The corresponding residual standard deviation is

$$\sigma_{C300} = 1.50343 \text{ ppm}. \quad (185)$$

1.26.5 B300 Mode

For the B300 mode,

$$f_{\text{ref},\text{B300}} = 10\,964\,730 \text{ Hz.} \quad (186)$$

The measured resonance frequencies are

$$\mathbf{f}_{\text{B300}} = [10967090, 10964730, 10963910, 10962210, 10960810, \\ 10959730, 10958190, 10956860, 10955510, 10954230] \text{ Hz.} \quad (187)$$

The fitted model is

$$\hat{p}_{\text{B300}}(T) = 672.7908 - 25.278849T \quad (188)$$

with temperature coefficient

$$b_{\text{B300}} = -25.278849 \frac{\text{ppm}}{^\circ\text{C}}. \quad (189)$$

The residuals are

$$\mathbf{r}_{\text{B300}} = [48.0217, -40.8196, 10.7894, -17.8589, -19.1468, \\ 8.7498, -5.3063, -0.2100, 3.0622, 12.7185] \text{ ppm.} \quad (190)$$

The residual sum of squares is

$$\text{SSE}_{\text{B300}} = 5050.1747 \text{ ppm}^2, \quad (191)$$

and therefore

$$\sigma_{\text{B300}}^2 = \frac{5050.1747}{8} = 631.2718 \text{ ppm}^2. \quad (192)$$

The corresponding residual standard deviation is

$$\sigma_{\text{B300}} = 25.1251 \text{ ppm.} \quad (193)$$

1.26.6 Summary of the Linearized Modes

The resulting linear temperature coefficients and residual variances are summarized in Table 1.

Table 1: Linear temperature coefficients and residual variances of the five resonant modes.

Mode	b [ppm/ $^\circ\text{C}$]	σ [ppm]	σ^2 [ppm 2]
C100	-1.171846	14.2234	202.3051
B100	-29.366872	69.5483	4836.9699
A100	-34.323755	52.8828	2796.5879
C300	-0.358644	1.5034	2.2603
B300	-25.278849	25.1251	631.2718

The corresponding temperature-coefficient vector, using the mode order

$$[C100, B100, A100, C300, B300],$$

is

$$\alpha = \begin{bmatrix} -1.171846 \\ -29.366872 \\ -34.323755 \\ -0.358644 \\ -25.278849 \end{bmatrix} \frac{\text{ppm}}{^\circ\text{C}}. \quad (194)$$

If the residual errors of the five modes are assumed to be statistically independent, the covariance matrix is diagonal and can be written as

$$\Sigma = \begin{bmatrix} 202.3051 & 0 & 0 & 0 & 0 \\ 0 & 4836.9699 & 0 & 0 & 0 \\ 0 & 0 & 2796.5879 & 0 & 0 \\ 0 & 0 & 0 & 2.2603 & 0 \\ 0 & 0 & 0 & 0 & 631.2718 \end{bmatrix} \text{ppm}^2. \quad (195)$$

These values represent the residual variance remaining after removal of the fitted linear temperature dependence. Consequently, they quantify the deviation of each measured mode from its linear temperature model, rather than the total frequency variation observed across the complete temperature sweep.

References

- [1] J. A. Kusters, M. C. Fischer, and J. G. Leach, “Dual mode operation of temperature and stress compensated crystals,” in *Proceedings of the Annual Symposium on Frequency Control*, pp. 389–397, 1978.